

[illegible]

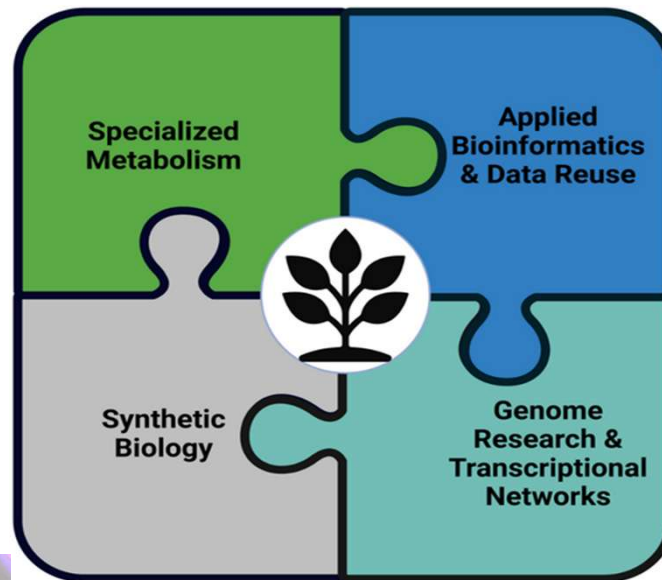
Samuel N. Meckoni & Nancy Choudhary

Command-line interface (CLI) basics for life scientists

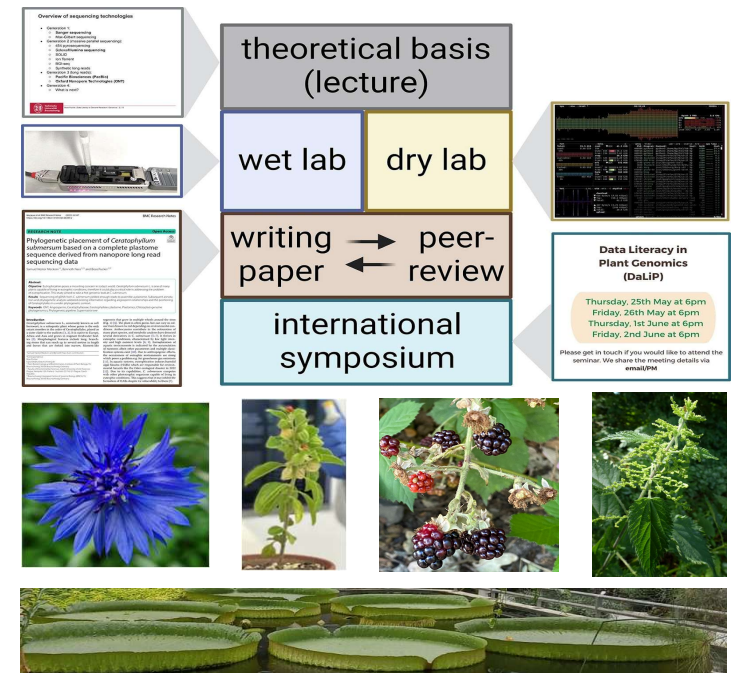
Plant Biotechnology & Bioinformatics (PBB)

Group overview

- Flavonoids:
Anthocyanins,
Flavonols, Flavones,
Proanthocyanidins
- Carotenoids
- Withanolides
- ...



Tools: KIPES, NAVIP, MGSE,
MYB annotator, DuplylCate...



Wolff *et al.*, 2023: 10.1515/jib-2023-0033 2

Dassow *et al.*, 2025: 10.1101/2025.01.31.635922

Course Overview

CLI basics for life scientists

- Basics of the Unix command line interface
- Unix file system
- Manipulating files and directories
- Combining bash commands
- Using these commands for biological insights



Samuel



Nancy



Boas

Availability of course material

- All materials will be shared at the end of this course.

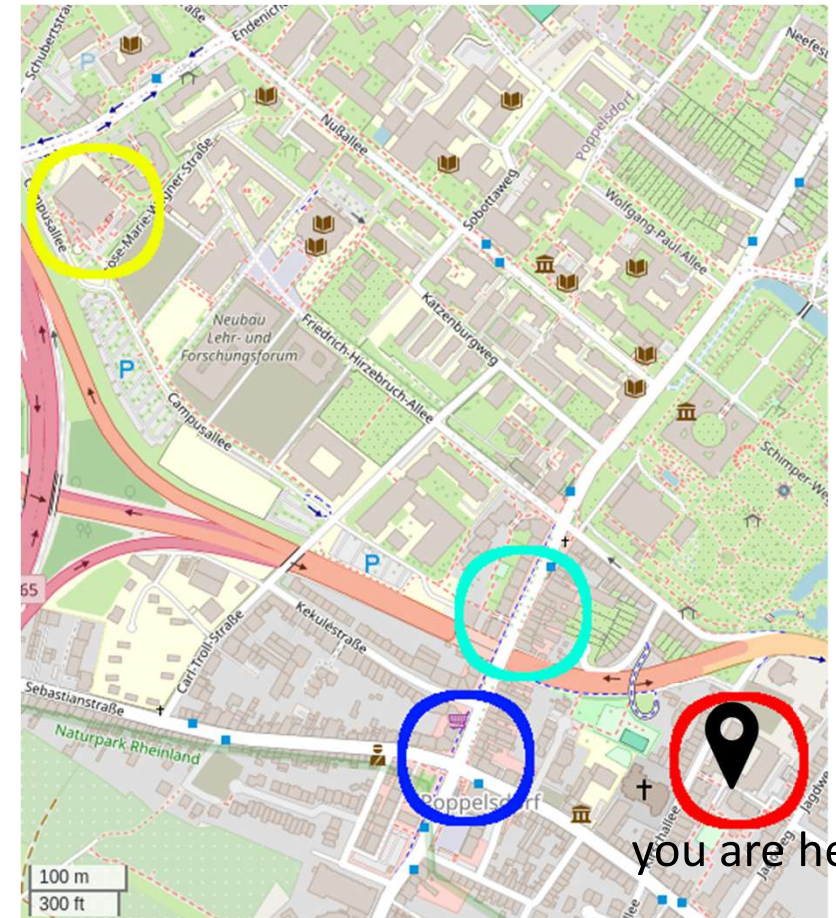
https://github.com/PuckerLab/teaching/tree/main/deNBI2026_CLIbasicsforlifescientists

Organisational I

- 10 AM -5 PM, 1 hour lunch break
- Tuesday: lunch break from 1 PM - 3 PM
 - PhD defense at 2 PM in 2.007
- Friday until 1 PM

Organisational II

- Mensa (15 mins walk)
- Local food stores
- Supermarkets



you are here

Recap

- SSH command → VM
- File system structure
- Basic commands

RTFM

RTFM

- Documentation = manual of a tool
 - Usually found on the internet

RTFM

- Documentation = manual of a tool
 - Usually found on the internet
- Inbuilt documentation
 - **man ls**
 - **cd --help**
 - **man -h**

RTFM

- Documentation = manual of a tool
 - Usually found on the internet
- Inbuilt documentation
 - **man ls**
 - **cd --help**
 - **man -h**
- (more help available on the internet, blog articles, community help, LLMs; **do not use any of these yet**)



Installing additional programs

apt = interface for the package management system

- apt update
- apt upgrade
- apt install [name of the tool]
- apt search [searchterm]
- apt-get moo?



Superuser right

- **sudo** (“super user do”)
- (or as root: **su -**)

Superuser right

- **sudo** (“super user do”)
- (or as root: **su -**)

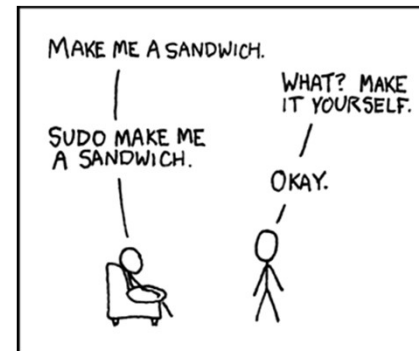


Figure “[Sandwich](#)” from “xkcd” is licensed under [CC BY-NC 2.5](#).

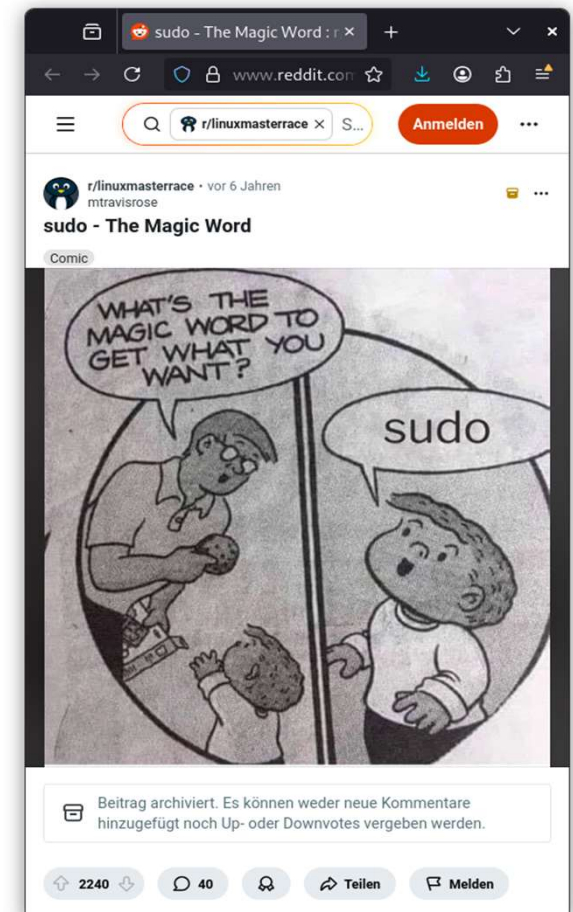


Figure: Screenshot from [reddit](#). Original figure source or creator could not be traced back further.

Shortcuts

- **Strg + c** = cancel a running progra,
- **Strg + shift + c** = copy selection
- **Strg + shift + v** = paste selection

Creating files

- **touch <filename>**
- **nano <filename>** (editor)

Common problems & solutions

- Check storage usage: **df, du, ncd**
- get full path of a file/directory : **realpath**
- Check permissions: **ls -l**

Task 2.1: Try out (play), read, and document

- You can find the commands here:
https://md.brothertec.eu/2026-09-21-CLI_course_tool_assignment#

Do not use LLMs!

Transferring files to and from VM

- **SCP (secure copy protocol)**
 - securely transfer files **between a local host and a remote host** or **between two remote hosts**
 - relies on SSH for data transfer
- **SSH**
 - ssh -i /path/to/your/ssh/private/key [ubuntu@194.94.113.18](#) -p 30XXX



Transferring files to and from VM

- **SCP (secure copy protocol)**

- securely transfer files **between a local host and a remote host** or **between two remote hosts**
- relies on SSH for data transfer

- **SSH**

```
ssh -i /path/to/your/ssh/private/key ubuntu@194.94.113.18 -p 30XXX
```

path to identity file

Transferring files to and from VM

- **SCP (secure copy protocol)**

- securely transfer files **between a local host and a remote host** or **between two remote hosts**
- relies on SSH for data transfer

- **SSH**

ssh -i  @194.94.113.18 -p 30XXX

path to identity file

Transferring files to and from VM

- **SCP (secure copy protocol)**

- securely transfer files **between a local host and a remote host** or **between two remote hosts**
- relies on SSH for data transfer

- **SSH**

```
ssh -i /path/to/your/ssh/private/key ubuntu@194.94.113.18 -p 30XXX
```

path to identity file IP address Port

Transferring files to and from VM

- **SCP (secure copy protocol)**

- securely transfer files **between a local host and a remote host** or **between two remote hosts**
- relies on SSH for data transfer

- **SSH**

```
ssh -i /path/to/your/ssh/private/key ubuntu@194.94.113.18 -p 30XXX
```

path to identity file IP address Port

- **Basic SCP command**

```
scp -P 3XXXX -i /path/to/your/ssh/private/key /source/file/folder/path  
ubuntu@194.94.113.18:/vol/data/destination/folder/
```



Transferring files to and from VM

- From a local host to a remote host

```
scp -P 3XXXX -i /path/to/your/ssh/private/key  
/source/file/folder/path  
ubuntu@194.94.113.18:/vol/data/destination/folder/
```

*The command is given from local machine (terminal)

Transferring files to and from VM

- From a local host to remote host

Fixed part

```
scp -P 3XXXX -i /path/to/your/ssh/private/key  
/source/file/path  
ubuntu@194.94.113.18:/vol/data/destination/folder/
```

*The command is given from local machine (terminal)

Two tricks:

- Info of the 'unknown' (IP address and port) is always required
- Order – what you want to transfer → where you want to transfer

Transferring files to and from VM

- From a remote host to local host

Fixed part

```
scp -P 3XXXX -i /path/to/your/ssh/private/key  
ubuntu@194.94.113.18:/vol/data/source/file  
/path/to/destination/folder/in/local/machine
```

*The command is given from local machine (terminal)

Two tricks:

- Info of the 'unknown' (IP address and port) is always required
- Order – what you want to transfer → where you want to transfer



Exercise

- **SSH into shared VM**

```
ssh -i /path/to/your/ssh/private/key ubuntu@194.94.113.18 -p 30197
```

Transfer the datasets folder from `/vol/data/datasets/` to your local machine, then from there to your VM.

*The command is given from local machine (terminal)

Two tricks:

- Info of the 'unknown' (IP address and port) is always required
- Order – what you want to transfer → where you want to transfer



Basic linux commands

Content representation

```
ls  
ls *.fasta  
ls [a-c]*.fasta
```

Content creation and removal

```
mkdir  
cp  
rm  
mv  
rmdir
```



Basic linux commands

Accessing content

more

less

head

tail

cat

wc or wc -l

Redirecting content

> (redirect output)

< (redirect input)

| (piping)



Linux commands

Querying content

sort
uniq
grep
cut

Redirecting content

> (redirect output)
< (redirect input)
| (piping)



Linux commands

Comparing content

diff

comm

Archiving content

gzip

gunzip

bzip2

bunzip2

tar -xvf



FASTA format

```
>ctg000000 type:s:linear length:i:316896 node:i:15
cgccgtccttggtcgacgaatctctggaccttcggtcgcatgaagttcctccagacctggtcgattttattttcaactttga
>ctg000010 type:s:linear length:i:18291631 node:s:unknown
cctaaacctaaacCCTAAACCTAAACCTAAACCTAAACCCTAAACCCTAAACCCTAAACCCTAAACCCTAAACCCTAAACC
>ctg000020 type:s:linear length:i:18476100 node:s:unknown
TCCTGCTTCTTTCGGCTTCGGAGGGCTGTCTTGGGCTTTCGAAAAGGTATCACATGCCAAGTTTGGCCTCACGGTCTAA
>ctg000030 type:s:linear length:i:32605930 node:i:1836
tgTAGGATCAATCATTGCATTTTCCATAGGATAGATAGGGCCGACAAGATCATCGGGGAAGAAGTCAAATCACATCCGAA
>ctg000040 type:s:linear length:i:11381128 node:i:703
cctaggcctaggcctaggcctgaaacctgaagcctaggcctaggcctgagcctaggcctaggcctgagcctgaagcctga
>ctg000050 type:s:linear length:i:324797 node:s:unknown
aatacatttttatttttttatttgacaagtactcgtactcaaaaaagttaaaaattttttattcatattttttttgtagtatg
```

GFF3 format

seqid	source	type	start	end	score	strand	phase	attributes
ctg1	GAF	gene	7031	7666	.	+	.	ID=gene001
ctg1	GeMoMa	mRNA	7031	7666	.	+	.	ID=gene001.1;Parent=gene001
ctg1	GeMoMa	CDS	7031	7097	.	+	0	Parent=gene001.1
ctg2	GeMoMa	CDS	7401	7666	.	+	2	Parent=gene001.1

Exercise

- Download *C. elegans* data from NCBI
GCF_000002985.6



NCBI RefSeq assembly GCF_000001735.4

Submitted GenBank assembly GCA_000001735.2

- Genome fasta (assembly)
- Annotation (gff3)
- Polypeptide seq (aa)

Assembly report

Index of /genomes/all/GC

Name

[Parent Directory](#)
[GCF_000002985.6_WBcel235_assembly_structure/](#)
[GCF_000002985.6_WBcel235_assembly_report.txt](#)
[GCF_000002985.6_WBcel235_assembly_stats.txt](#)
[GCF_000002985.6_WBcel235_cds_from_genomic.fna.gz](#)
[GCF_000002985.6_WBcel235_fcs_report.txt](#)
[GCF_000002985.6_WBcel235_feature_count.txt](#)
[GCF_000002985.6_WBcel235_feature_table.txt.gz](#)
[GCF_000002985.6_WBcel235_genomic.fna.gz](#)
[GCF_000002985.6_WBcel235_genomic.gbff.gz](#)
[GCF_000002985.6_WBcel235_genomic.gff.gz](#)
[GCF_000002985.6_WBcel235_genomic.gtf.gz](#)
[GCF_000002985.6_WBcel235_protein.faa.gz](#)
[GCF_000002985.6_WBcel235_protein.gpff.gz](#)
[GCF_000002985.6_WBcel235_rna.fna.gz](#)
[GCF_000002985.6_WBcel235_rna.gbff.gz](#)
[GCF_000002985.6_WBcel235_rna_from_genomic.fna.gz](#)
[GCF_000002985.6_WBcel235_translated_cds.faa.gz](#)
[README.txt](#)
[annotation_hashes.txt](#)
[assembly_status.txt](#)
[md5checksums.txt](#)
[uncompressed_checksums.txt](#)

Exercise

- How many sequences are there in each genome fasta file (calculate from .genome.fasta)?
- How many sequences are there in each genome sequence (calculate from gff3)?
- How many number of genes and transcripts?
- How many genes are there on each of the '+' and '-' strands?

What we will learn tomorrow?

- Sequences and genomic features
- Tools for transcriptomics

